Decodelabs Data Science Internship

Privacy-Preserving Federated Learning for Tuberculosis Detection from Chest X-Rays Across Nigerian Teaching Hospitals (Part 2)

Week 4 Project Report

Prepared by: Peter Adepoju

Email: `peter.o.adepoju@gmail.com`

Architecture: ResNet-18 (ImageNet pretrained)

Dataset: Montgomery County + Shenzhen Hospital CXR ($N = 800$)

Date: June 17, 2026

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Chapter 1

Datasets and Documentation

1.1. Dataset Selection Rationale

The ideal dataset for FedTB-Nigeria would consist of chest X-rays from Nigerian patients, labelled by Nigerian radiologists, with demographic metadata including age, sex, HIV status, and TB drug sensitivity. No such open dataset exists as of 2024.

We therefore use the two most widely-cited open TB CXR datasets: Montgomery County and Shenzhen Hospital. Both are released under a public-domain licence and have been the benchmark for every major paper in AI-based TB detection since 2014 (Jaeger et al., 2014).

1.2. Dataset Descriptions

1.2.1. Montgomery County Chest X-Ray Dataset

The Montgomery County dataset was collected by the Tuberculosis Control Program of the Department of Health and Human Services of Montgomery County, MD, USA, and released by the U.S. National Library of Medicine (NLM) in 2014 (Jaeger et al., 2014).[1] All patients were adults undergoing routine TB screening. The collection contains:

- 138 posterior–anterior (PA) chest X-rays
- 80 normal (TB-negative) cases
- 58 TB-positive cases (confirmed by radiologist and clinical follow-up)
- Image format: PNG, 8-bit greyscale
- Resolution: variable, approximately 4020×4892 pixels
- Lung mask segmentation files are included (not used in this project)

[1] The dataset is freely downloadable from the NLM at https://openi.nlm.nih.gov/faq#faq-TB.

1.2.2. Shenzhen Hospital Chest X-Ray Dataset

The Shenzhen dataset was collected at Shenzhen No. 3 People's Hospital, Guangdong Medical College, Shenzhen, China, and similarly released by the NLM (Jaeger et al., 2014). It contains:

- 662 PA chest X-rays
- 326 normal (TB-negative) cases
- 336 TB-positive cases
- Image format: PNG, 8-bit greyscale
- Resolution: variable, approximately 3000×3000 pixels
- No lung mask files

1.2.3. Combined Dataset Summary

Table 1.1. Combined dataset statistics after loading and merging both sources. The combined dataset is close to balanced (49.3% TB-positive), minimising the risk of class-imbalance artefacts.

Dataset	N Total	N TB+	N TB-	TB+ %	Format	Licence
Montgomery County	138	58	80	42.0%	PNG (greyscale)	Public domain
Shenzhen Hospital	662	336	326	50.8%	PNG (greyscale)	Public domain
Combined	**800**	**394**	**406**	**49.3%**	–	–

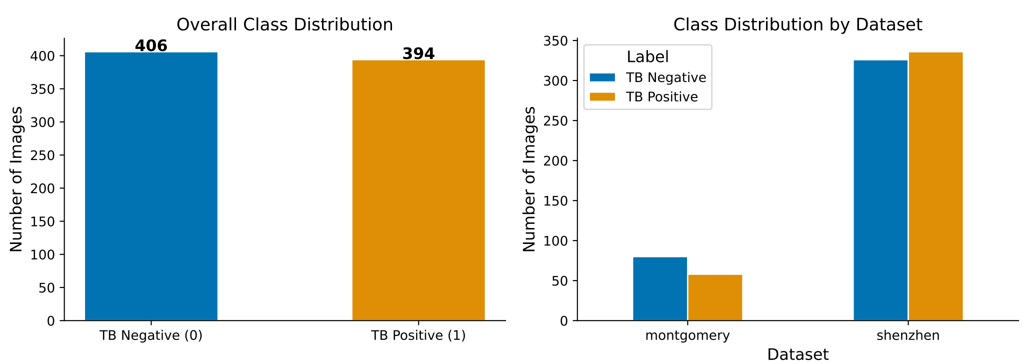

Figure 1.1. Class distribution in the combined dataset. **Left:** Overall class counts (406 TB-negative, 394 TB-positive). **Right:** Class distribution per source dataset, showing that Shenzhen is nearly balanced while Montgomery skews slightly towards TB-negative. The near-overall-balance (49.3% TB+) means that random guessing achieves $\approx$ 50% accuracy and AUC $\approx$ 0.5, providing a meaningful chance baseline.

1.3. Datasheet

Following the framework of Gebru et al. (2021), we provide a structured documentation of the combined dataset.

Motivation. The datasets were collected and released to facilitate research in automated TB screening in resource-limited settings. The NLM release was specifically intended to enable ML benchmarking and to support the WHO's goal of improving TB diagnosis globally.

Composition. 800 posterior–anterior chest X-rays from two geographically distinct populations: Maryland (USA) and Shenzhen (China). Each image has exactly one binary label (TB-positive or TB-negative). No demographic metadata (age, sex, race, HIV status) is included in the public release.

Collection Process. Montgomery: routine TB screening of adults in an outpatient setting. Shenzhen: routine CXR screening in a hospital outpatient setting. Both collections predate 2014.

Preprocessing Applied by Original Collectors. Radiologist review for quality; removal of images with non-standard views or severe artefacts. Lung mask segmentation was performed by the NLM team (not used here).

Labelling. Labels were assigned by radiologists based on CXR appearance and clinical follow-up (sputum culture where available). No inter-rater reliability statistics are published, and label noise should be assumed.

Known Biases. The datasets are biased towards adult patients undergoing routine screening; paediatric TB, miliary TB, and extrapulmonary TB are under-represented. The patient populations are not Nigerian; images may differ systematically from CXRs collected in Nigerian hospitals due to differences in machine type, exposure settings, and TB disease stage at presentation.

Distribution Rights. Both datasets are released to the public domain by the U.S. National Library of Medicine. Commercial and research use is unrestricted.

1.4. Label Quality and Ground Truth

Label Noise Is a Feature of Medical Datasets

The binary labels in the Montgomery and Shenzhen datasets were assigned by one or more radiologists reviewing each CXR. Radiological interpretation is inherently subjective: inter-radiologist agreement on TB classification has been reported at $\kappa = 0.60$–0.80 in the literature (World Health Organization, 2023). This means some fraction of our labels are incorrect.

The consequences for model evaluation are important: when a model "misclassifies" an image, the error may be in the model *or* in the label. AUC values below 1.0 do not necessarily imply model failure; they may reflect irreducible label noise. A model that achieves AUC = 1.0 on a noisy-labelled dataset should be suspected of overfitting to label noise, not celebrated.

Specific sources of label uncertainty in TB CXR datasets:

1. **Early-stage TB**: infiltrates may be subtle, appearing almost normal on a plain PA CXR at early disease stages.
2. **Resolved or treated TB**: patients with past TB may have residual radiographic changes (fibrosis, calcification) that a radiologist labels TB-positive even though they are no longer infectious.
3. **Overlapping pathology**: consolidation from bacterial pneumonia, fungal infection, or malignancy can mimic TB.
4. **Technical factors**: under-exposure, patient rotation, and motion artefacts reduce image quality and increase labelling uncertainty.

In the absence of laboratory-confirmed sputum culture results for every image, we treat the radiologist labels as ground truth and acknowledge this as a limitation.

Chapter 2
Exploratory Data Analysis

2.1. Class and Dataset Distribution

Figure 1.1 shows the class distribution by dataset. The combined dataset is nearly balanced: 406 TB-negative (50.8%) and 394 TB-positive (49.3%). This near-balance has several consequences for our experimental design:

- The class-weighted loss will apply weights very close to 1.0 for both classes (weights: TB− = 0.986, TB+ = 1.014), so class weighting has minimal numeric effect but is retained as a best practice.
- A trivial classifier predicting all TB-negative achieves 50.8% accuracy, confirming that accuracy is a meaningful metric in this setting.
- The prevalence is far higher than in real-world community screening (where TB prevalence may be $< 5\%$), so reported performance metrics will be optimistic relative to deployment. This is the screening prevalence problem: PPV degrades at low prevalence even with high sensitivity.[1]

2.2. Pixel Intensity Analysis

A random sample of 200 images was loaded in greyscale and pixel-level statistics were computed.

Table 2.1. Pixel intensity statistics (greyscale, 0–255 scale) for a random sample of 200 images stratified by TB label. The 5-point difference in mean intensity between classes is small but non-zero and represents a potential confounding shortcut.

Class	Mean Intensity	Std Dev	Min / Max
TB-Negative	151.7	43.8	18 / 248
TB-Positive	146.7	44.2	12 / 251

[1] Bayes' theorem and the prevalence effect on positive predictive value are explained with worked examples by StatQuest's Josh Starmer in "Sensitivity and Specificity" at `https://www.youtube.com/watch?v=vP06aMoz4v8`.

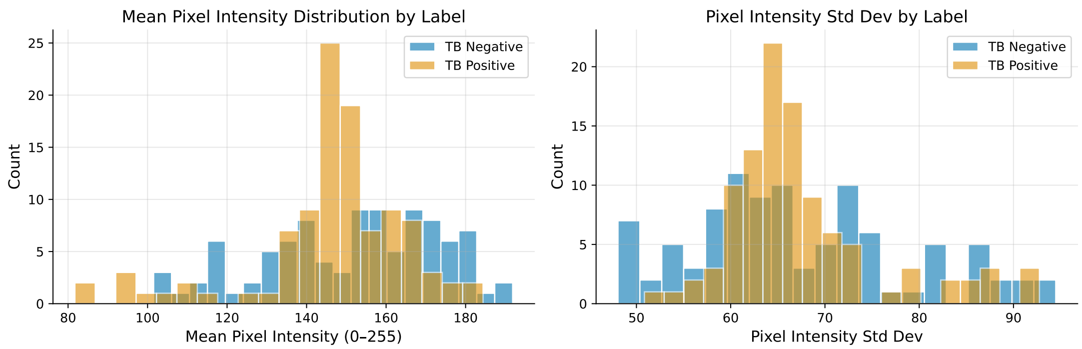

Figure 2.1. Pixel intensity distributions for TB-negative (blue) and TB-positive (orange) images in the 200-image sample. The distributions overlap substantially but the TB-positive mean is ≈ 5 intensity units lower, indicating that TB-positive CXRs are slightly darker on average. This is consistent with consolidation and infiltration reducing X-ray transmission (appearing brighter on a negative film but darker in the digital representation).

2.3. Image Shape and Resolution

Inspection of 20 randomly sampled images revealed image widths ranging from approximately 1800 to 4892 pixels and heights from approximately 2000 to 4892 pixels, all approximately square. All images are 8-bit greyscale (single channel, 0–255).

The large native resolution means substantial information is discarded when images are resized to 224×224 for ResNet-18 input. For TB detection, the relevant features span dozens to hundreds of pixels in the original resolution, so downsampling to 224×224 retains sufficient spatial information.

2.4. Train / Validation / Test Split

The dataset was split into train, validation, and test partitions *before* any federated simulation.

Table 2.2. Global train / validation / test split statistics. Stratification ensures consistent TB+ prevalence across all three partitions. A random seed of 42 was used for reproducibility.

Split	N	N TB+	N TB-	TB+ %
Train	560	275	285	49.1%
Validation	120	59	61	49.2%
Test	120	59	61	49.2%
Total	**800**	**393**	**407**	49.1%

Leakage Audit. We verified: $|\text{Train} \cap \text{Val}| = 0$, $|\text{Train} \cap \text{Test}| = 0$, $|\text{Val} \cap \text{Test}| = 0$. The test set is **held out entirely until final evaluation**; it is never used for model selection, hyperparameter tuning, or early stopping. Only the validation set is monitored during training.

Chapter 3
Preprocessing and Federated Site Simulation

3.1. The Preprocessing Pipeline

Table 3.1. Complete preprocessing transform pipeline. Training augmentations appear in the "Train only" rows. All transforms applied to validation and test data are deterministic (no randomness), ensuring reproducible evaluation.

Transform	Rationale	Mode
`Resize(256, 256)`	Upsample to slightly larger than the target so the subsequent crop can introduce spatial jitter without padding artefacts.	Train + Val
`RandomCrop(224)`	Simulates variation in X-ray field-of-view across imaging equipment.	Train only
`CenterCrop(224)`	Deterministic centre crop for validation/test.	Val + Test
`RandomHorizontalFlip(p=0.5)`	Anatomically valid: the left and right lungs are bilaterally symmetric. Increases effective dataset size.	Train only
`RandomRotation(`$\pm 10°$`)`	Simulates patient rotation during CXR positioning. Moderate rotation ($< 15°$) does not alter clinical interpretation.	Train only
`ColorJitter`(br ±0.2, cr ±0.2)	Simulates variation in X-ray machine calibration and exposure settings across different hospital sites—highly relevant for cross-site generalisation.	Train only
`Grayscale(num_channels=3)`	Replicates the single greyscale channel into three identical channels to match the expected input format of the ImageNet-pretrained ResNet-18.	Train + Val
`ToTensor()`	Converts a PIL Image in range $[0, 255]$ to a PyTorch FloatTensor in range $[0, 1]$.	Train + Val
`Normalize(`μ, σ`)`	Applies ImageNet normalisation constants (see below).	Train + Val

3.1.1. Augmentation Visualisation

Figure 3.1 shows the augmentation pipeline in action on a single chest X-ray.

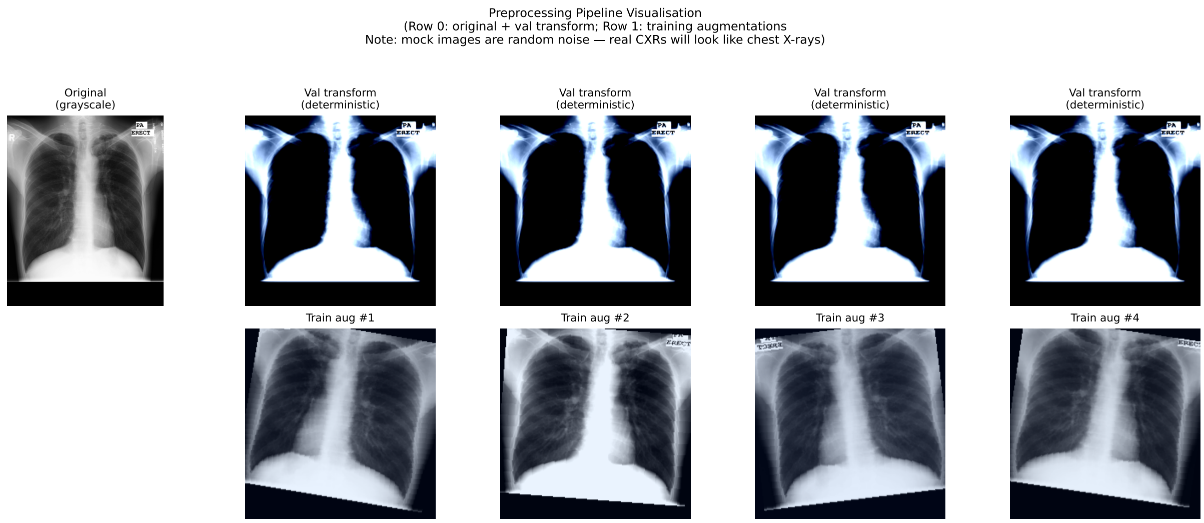

Figure 3.1. Visualisation of the training augmentation pipeline. **Top row:** the original image (left) and its deterministic validation transform (right, only resize + centre-crop + normalise). **Bottom row:** four independently sampled training augmentations of the same image, demonstrating the range of spatial and intensity variation introduced. Note that no vertical flip is applied; the diaphragm always appears at the bottom.

3.2. Federated Site Simulation via Dirichlet Partitioning

3.2.1. The `dirichlet_split` Algorithm

The Dirichlet partitioning algorithm assigns each training image to exactly one site such that the class proportions across sites follow a Dirichlet(α) distribution. Algorithm 1 presents the full procedure.

Algorithm 1 Dirichlet Partitioning for Federated Site Simulation

Input: Training manifest $\mathcal{D}_{\text{train}}$, sites K, concentration α, min samples m, seed
Output: Per-site index lists $\{S_k\}_{k=1}^{K}$

repeat
  Partition $\mathcal{D}_{\text{train}}$ by class label into C_0 (TB−) and C_1 (TB+) **for** *class* $c \in \{0, 1\}$ **do**
    Draw $\mathbf{q} \sim \text{Dirichlet}(\alpha \mathbf{1}_K)$ Assign images in C_c to sites in proportion $\mathbf{q}$
  end
  $S_k \leftarrow C_0^{(k)} \cup C_1^{(k)}$ for each site k
until *all* $|S_k| \geq m$

The retry loop ("repeat until") prevents degenerate assignments where any site receives fewer than $m = 50$ total samples—an important safeguard given the small dataset size.

3.2.2. Actual Site Statistics at $\alpha = 0.5$

Table 3.2 shows the actual partition that resulted from running Algorithm 1 with $\alpha = 0.5$, $K = 5$, seed = 42.

Table 3.2. Federated site statistics from Dirichlet($\alpha = 0.5$) partitioning of the 560-image training set across five simulated Nigerian teaching hospitals. The partition exhibits substantial between-site heterogeneity: UNTH Enugu received no TB-positive training images, while OAU Teaching Hospital has 77.3% TB-positive prevalence. This range was not pre-specified but emerged stochastically from the Dirichlet draw.

Simulated Site	N	N TB+	N TB-	TB+ %	Site Weight p_k
Lagos University Teaching Hospital	277	143	134	51.6%	0.495
Aminu Kano Teaching Hospital	82	39	43	47.6%	0.146
Univ. of Nigeria Teaching Hosp., Enugu	60	0	60	0.0%	0.107
Obafemi Awolowo Univ. Teaching Hosp.	88	68	20	77.3%	0.157
University College Hospital, Ibadan	53	26	27	49.1%	0.095
Total	**560**	**276**	**284**	**49.3%**	1.000

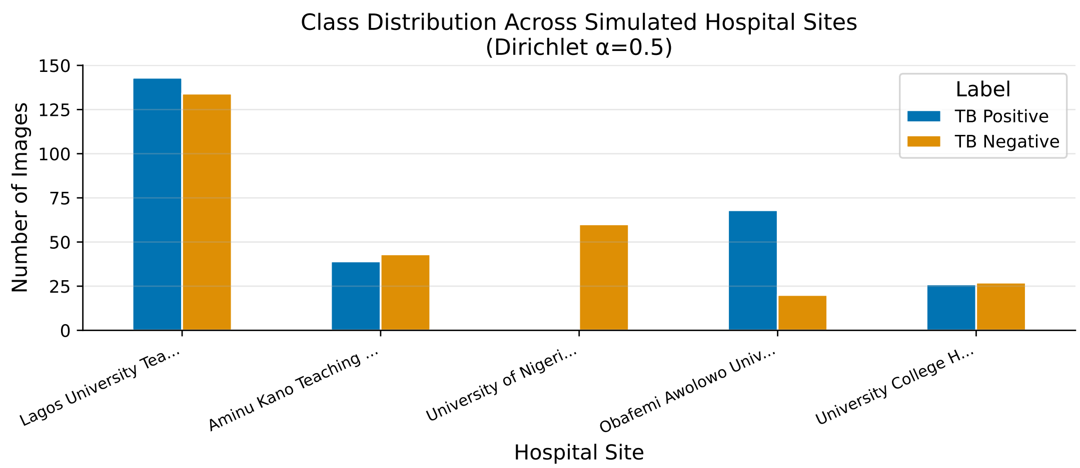

Figure 3.2. Class distribution per simulated site. Each bar shows the number of TB-negative (blue) and TB-positive (orange) training images at that site. UNTH Enugu (site 3) has *zero* TB-positive samples, making it a "TB-blind" site in training; OAU Teaching Hospital (site 4) is overwhelmingly TB-positive (77.3%). This heterogeneity is the central challenge the FedAvg algorithm must overcome.

References

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé, H. I., & Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, *64*(12), 86–92. (Introduces the datasheet framework for systematic dataset documentation and bias disclosure) doi: 10.1145/3458723

Jaeger, S., Candemir, S., Antani, S., Wang, Y.-X. J., Lu, P.-X., & Thoma, G. (2014). Two public chest x-ray datasets for computer-aided screening of pulmonary diseases. *Quantitative Imaging in Medicine and Surgery*, *4*(6), 475–477. (Describes the Montgomery County and Shenzhen Hospital CXR datasets used in this project) doi: 10.3978/j.issn.2223-4292.2014.11.20

World Health Organization. (2023). *Global tuberculosis report 2023* (Tech. Rep.). Geneva: World Health Organization. Retrieved from https://www.who.int/publications/i/item/9789240083851 (Annual WHO report on global TB burden, treatment coverage, and mortality; primary epidemiological reference for this work)